

Differentiated Bertrand

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Grad IO

Multiproduct Demand system estimation is probably the most important contribution from the New Empirical IO literature

- ▶ Demand is an important primitive (think about Econ 101)
- ▶ Welfare Analysis, Pass through of taxation, Value of Advertising, Price Effects of Mergers all rely on demand estimates.
- ▶ Last 5-10 years has seen successful export to other fields: Trade, Healthcare, Education, Urban Economics, Marketing, Operations Research.
- ▶ Anywhere consumers face a number of options and “prices”: doctors, hospitals, schools, mutual funds, potential dates, etc.
- ▶ There are many many more applications.

► Endogeneity of Prices

- Prices are not randomly determined, but set strategically by firms who observe the demand curve — this is exactly the **simultaneity** problem of Working (1927) from Week 1.

► Multiple Products/Flexibility

- We want to allow for flexible (data driven) substitution across products but if we have J products, then $\partial Q_j / \partial P_k$ might have J^2 elements.
- We may also think that $\partial Q_j / \partial P_k$ varies with P and with other covariates x .
- We may also care about $\partial^2 Q_j / \partial P_k^2$, $\partial^3 Q_j / \partial P_k^3$ and so on.

► We will address the issues separately and then see how to put them back together.

- ▶ Representative Consumer vs. Heterogeneous Agents?
- ▶ Discrete Choices vs. Continuous Choices?
- ▶ Single Product vs. Many Products?
- ▶ Product Space vs. Characteristic Space?
 - Do consumers choose products in product space or in characteristic space?

Data Sources

- ▶ We can either have **aggregate data** (market level data on (P_j, Q_j))
 - Many supermarket “scanner” datasets: Nielsen, IRI, Dominick’s
 - NPD: video games/computers/consumer electronics.
 - Many proprietary single firm sources
- ▶ or **micro data** panel of data with same individuals over time.
 - Best example is Nielsen Homescan Consumer Panel data.
 - Visa/Mastercard datasets
 - Medicare
- ▶ Sometimes we have a combination of both
 - Often we have aggregate purchase data plus some **micro data** from a survey on a subpopulation.
 - ie: asking people who purchased GM cars which other cars they were considering.
 - Using scanner data for alcohol purchases and comparing to consumption surveys by income and education.

Differentiated Products Bertrand

Consider the multi-product Bertrand problem where firms solve:

$$\arg \max_{p \in \mathcal{J}_f} \pi_f(\mathbf{p}) = \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p}):$$

$$0 = q_j(\mathbf{p}) + \sum_{k \in \mathcal{J}_f} (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})$$

$$\rightarrow p_j = q_j(\mathbf{p}) \left[-\frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right]^{-1} + c_j + \underbrace{\sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p}) \left[-\frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right]^{-1}}_{D_{jk}(\mathbf{p})}$$

$$p_j(p_{-j}) = \underbrace{\frac{1}{1 + 1/\epsilon_{jj}(\mathbf{p})}}_{\text{Markup}} \left[c_j + \underbrace{\sum_{k \in \mathcal{J}_f \setminus j} (p_k - c_k) \cdot D_{jk}(\mathbf{p})}_{\text{opportunity cost}} \right].$$

We call $D_{jk}(\mathbf{p}) = \frac{\frac{\partial q_k}{\partial p_j}(\mathbf{p})}{\left| \frac{\partial q_j}{\partial p_j}(\mathbf{p}) \right|}$ the **diversion ratio**.

It is helpful to define the matrix Δ with entries:

$$\Delta_{(j,k)}(\mathbf{p}) = \begin{cases} -\frac{\partial q_j}{\partial p_k}(\mathbf{p}) & \text{for } (j,k) \in \mathcal{J}_f \\ 0 & \text{for } (j,k) \notin \mathcal{J}_f \end{cases}$$

We can re-write the FOC in matrix form:

$$q(\mathbf{p}) = \Delta(\mathbf{p}) \cdot (\mathbf{p} - \mathbf{mc})$$

And recover markups and marginal costs:

$$(\mathbf{p} - \mathbf{mc}) = \Delta(\mathbf{p})^{-1} q(\mathbf{p}) \Rightarrow \mathbf{mc} = \mathbf{p} - \Delta(\mathbf{p})^{-1} q(\mathbf{p})$$

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What do we want to learn from demand systems?

- ▶ We need two objects from the demand system:
 - The (vector of) **predicted sales** at the (vector of) prices $\mathbf{q}(\mathbf{p})$.
 - The **derivatives** (or elasticities) of the demand function $\frac{\partial q_k}{\partial p_j}(\mathbf{p})$
 - Or the **diversion ratios**: $D_{jk}(\mathbf{p})$.
 - Welfare is often related to substitution to no purchase or “outside goods”.
- ▶ Ideally, these objects depend on the full vector of prices $\mathbf{p}$.
 - We can impose restrictions on the demand curve, such as constant slope or constant elasticity.
- ▶ With these in hand, we can:
 - Recover estimates of marginal cost (remember these are proprietary firm information and accounting estimates are often unreliable).